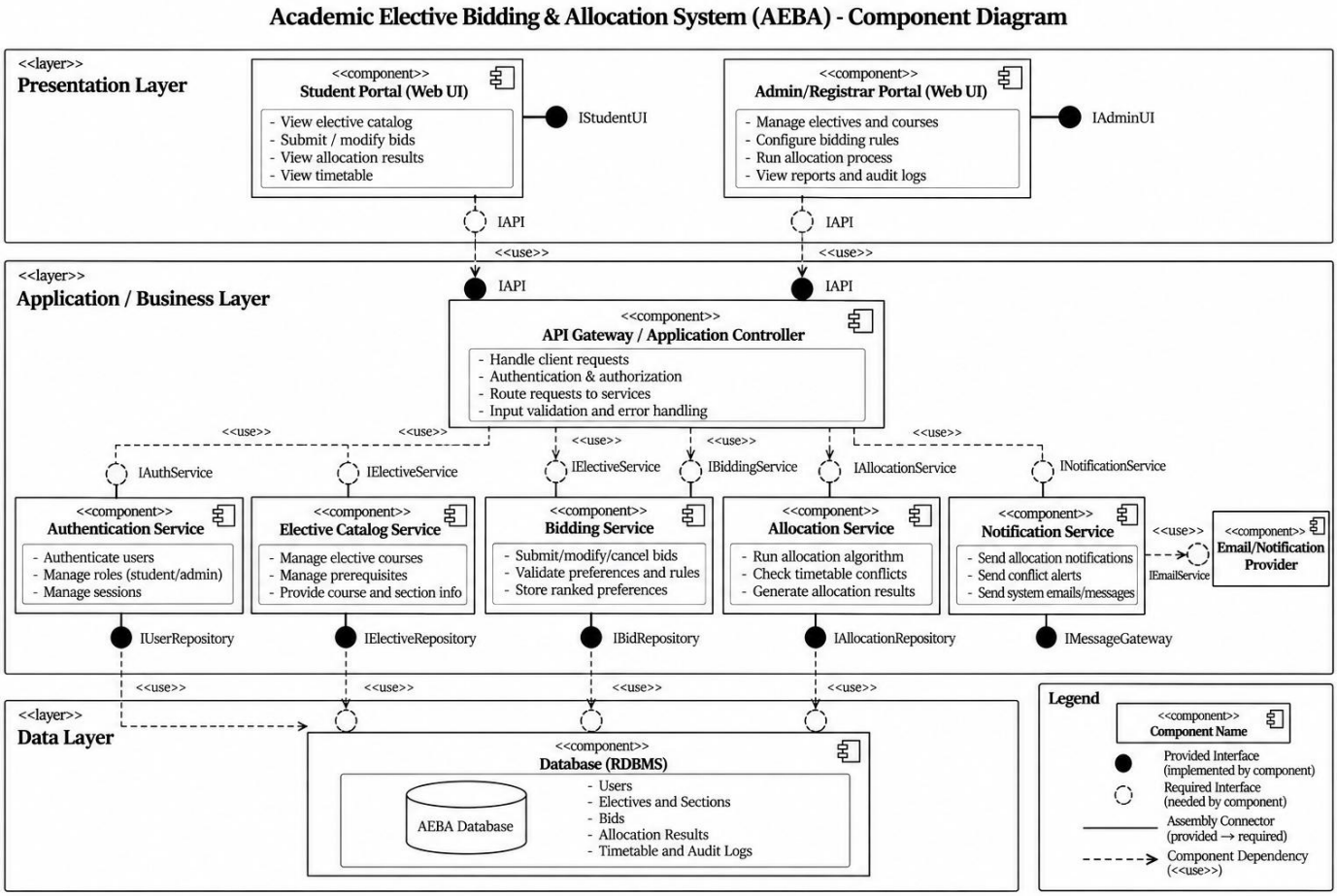
Lab 3: Component Modelling & Architectural Pattern

Team 05: Academic Elective Bidding and Allocation System



Architecture Selection and Justification

System: Academic Elective Bidding & Allocation System (AEBA)
Selected Architectural Pattern: Layered Architecture

1. Architectural Choice

We selected **Layered Architecture** for the Academic Elective Bidding & Allocation System. The system is organized into three layers: Presentation Layer, Application/Business Logic Layer, and Data Layer.

The Presentation Layer provides interfaces for students and academic registrars. The Business Logic Layer manages authentication, elective catalogs, student bidding, allocation processing, timetable conflict checking, and

notifications. The Data Layer manages persistent storage of student information, elective details, bids, allocation results, and audit records.

2. Reasons for Selection

Reason 1 — Separation of Concerns: Layered Architecture separates user interface operations, business rules, and data management. This makes the elective bidding and allocation workflow easier to understand, develop, test, and maintain. Changes to the student portal can be made without directly modifying the allocation algorithm or database structure.

Reason 2 — Modularity and Maintainability: Functional components such as Bidding Service, Elective Catalog Service, and Allocation Service have clearly defined responsibilities. This allows changes to allocation rules, prerequisites, seat capacities, and timetable conflict validation without redesigning the entire application.

3. Security Advantage

Layered Architecture supports centralized authentication, authorization, and input validation in the application layer. Role-based access control can restrict students to submitting bids and viewing their own results, while allowing authorized registrars to manage electives and execute allocation processes. Database access is restricted through the business layer rather than exposed directly to users.

4. Performance Benefit

Separating the presentation, business, and data layers allows the system to optimize database queries and reuse frequently accessed elective catalog information through caching. The allocation engine can process student preferences and seat availability centrally, reducing unnecessary data transfers between the user interface and database.

Conclusion

Layered Architecture is suitable for AEBA because it provides clear separation of responsibilities, modularity, controlled data access, and opportunities for performance optimization while keeping the system manageable for academic development.